

Aayan Mapara

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EDUCATION

Arizona State University

Tempe, AZ

Bachelor of Science in Computer Science, Minor in Mathematics

Aug 2025 – Dec 2028

4.0/4.0 GPA | Data Structures & Algorithms, Programming Languages, Calculus III, Linear Algebra, Discrete Math

EXPERIENCE

Undergraduate Researcher

Sep 2026 – Present

Make Programming Simple Lab

Tempe, AZ

- Developing a PyTorch compiler pass that selects between FP64 and Ozaki Tensor Core emulation for matmuls
- Building accuracy and cost models to select emulation strategy, slice count, and backend kernel per workload
- Integrating compiler routing into torch.compile so high-precision matmuls dispatch to optimized backend kernels
- Benchmarking runtime and error against native FP64 and existing emulation methods across ML and HPC tasks

Data Acquisition Engineer

Sep 2025 – Present

Sun Devil Motorsports Formula SAE

Tempe, AZ

- Wrote a Python lap simulator based on real car geometry and thermal tire models to generate a telemetry dataset
- Trained and quantized a TensorFlow tire-slip ratio model to 0.004 MAE and deployed it on STM32 at 5 ms latency
- Wrote ESP32 firmware that monitors coolant flow rate and temperature, transmitting over CAN every 100 ms
- Built an HTTP file server in C on an ESP32 to reduce log retrieval time by 30 minutes for 15 trackside engineers

PROJECTS

Recon | Python, JavaScript, React, Pandas, Scikit-Learn, AWS, Docker | [Repo](#) | [App](#)

Oct 2025 – Feb 2026

- Deployed a containerized FastAPI inference service to AWS EC2 that samples 14 Ethereum addresses per minute
- Engineered a Pandas pipeline that computes 22 Ethereum and 792 ERC-20 features per sampled address
- Trained a random forest classifier on 814 transaction features to flag fraud at 0.99 precision and 0.83 recall
- Built a React dashboard that streams scored transactions and flags high-risk addresses during block processing

ScribeShield: Medical STT App | TypeScript, FastAPI, Next.js, PyTorch, Git | [Repo](#)

Apr 2026

- Built a hackathon-winning transcription system to improve accuracy on telephony audio in telemedicine workflows
- Developed a Next.js web app that records and streams live consultation audio and produces tailored care plans
- Implemented LoRA fine-tuning on a Whisper Small model, reducing word error rate on medical terms to 4.14%
- Created a FastAPI service serving 3 fine-tuned STT model variants for live inference and transcription

CUDA GPU Kernels: FlashAttention-2 & SGEMM | C++, CUDA | [Repo](#)

May 2026 – Jun 2026

- Hit 90.94% of cuBLAS on a 3060Ti with a hand-written SGEMM kernel using 2D block tiling and vectorized loads
- Accelerated matrix multiplication 61x by autotuning tile dimensions and implementing shared-memory reuse
- Built a FlashAttention-2 forward kernel from scratch using tiled attention and stable online softmax rescaling
- Raised SM throughput from 30.81% to 59.26% and cut warp latency from 11.1 to 4.34 cycles with double buffering

LEADERSHIP

Technology Officer

May 2026 – Present

Software Developers Association @ ASU

Tempe, AZ

- Organize and host technical workshops covering topics such as agentic coding tools for 200+ ASU students
- Develop and maintain the public website, CSWiki, and tools used by 20+ officers and 500 students

AWARDS & ACHIEVEMENTS

Hackathons: 1st Place Innovation Hacks 2026, 1st Place VillageHacks 2026

Awards: National Merit Scholar, NVIDIA Deep Learning Institute AI & ML Certificate of Competency

TECHNICAL SKILLS

Languages: Python, C, C++, Rust, Java, TypeScript, SQL

Infrastructure & Frameworks: Git, AWS, Docker, CI/CD, REST, FastAPI, React, Next.js

AI & ML: PyTorch, TensorFlow, CUDA, HuggingFace, Scikit-learn, LangGraph, Pandas, NumPy

Embedded & Hardware: ESP32, STM32, ROS2, Jetson